

Phase Interpolators

From I/Q First Principles to SerDes & Beam Steering

Rahul Narwar

HIE Lab, University of California, Irvine

2026-04-24

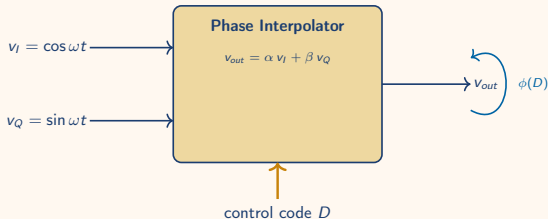
First Principles

What is a Phase Interpolator?

A circuit that produces a clock at **any phase between two reference phases** by taking a weighted sum.

- **Inputs:** two reference clocks, same frequency, known phase offset (typically 90° apart — I and Q)
- **Control:** digital code $\rightarrow$ analog weights (α, β)
- **Output:** one clock with phase $\phi \in [\phi_I, \phi_Q]$, same frequency

No frequency synthesis, no PLL — *just controlled mixing of two phases.*



Black box: 2 phases in, 1 phase out, code controls where.

Given two orthogonal clocks at ω :

$$v_I(t) = A \cos \omega t, \quad v_Q(t) = A \sin \omega t.$$

Take a weighted sum with real weights α, β :

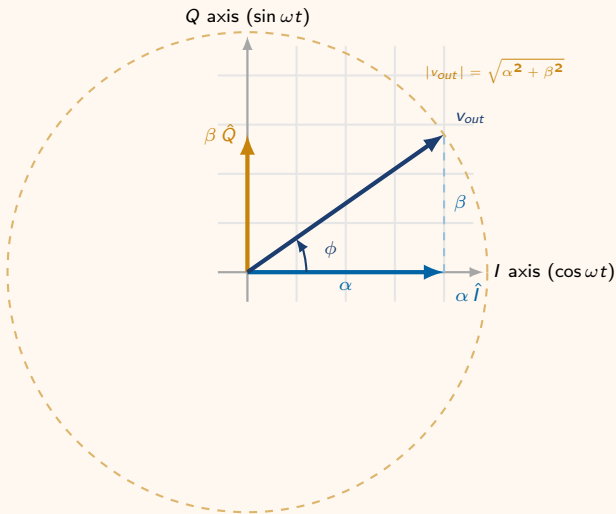
$$v_{out}(t) = \alpha v_I(t) + \beta v_Q(t) = A[\alpha \cos \omega t + \beta \sin \omega t].$$

Apply the harmonic-addition identity $a \cos \theta + b \sin \theta = R \cos(\theta - \phi)$ with $R = \sqrt{a^2 + b^2}$, $\tan \phi = b/a$:

$$v_{out}(t) = A \sqrt{\alpha^2 + \beta^2} \cos(\omega t - \phi), \quad \phi = \arctan \frac{\beta}{\alpha}.$$

Takeaway

The output *frequency* is unchanged. The output *phase* is set entirely by the weight ratio β/α — sweep the weights, sweep the phase from 0° (pure I) to 90° (pure Q).

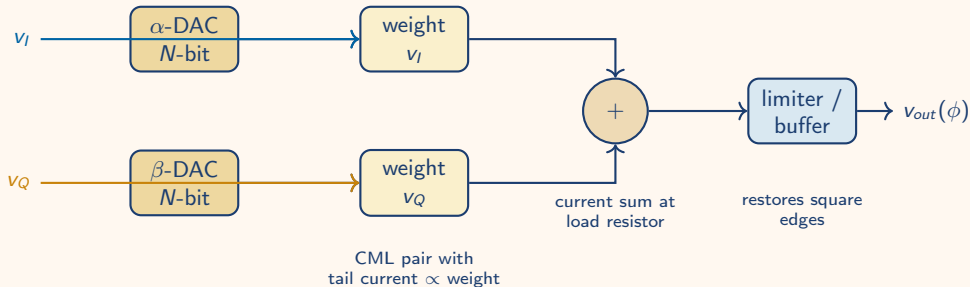


Geometry = algebra:

- α slides the tip along I
- β slides the tip along Q
- Tip's polar angle is ϕ

Two knobs:

- **Quadrant select** (coarse) — pick two of $\{I, Q, \bar{I}, \bar{Q}\}$ for 360° coverage
- **Weight ratio** (fine) — N -bit DAC on β/α gives 2^N steps per quadrant



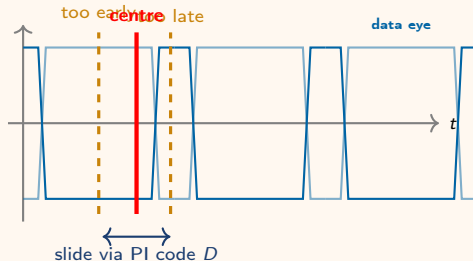
Two CML (current-mode logic) differential pairs share a load. Tail-current ratio $I_\alpha : I_\beta$ is the weight ratio $\alpha : \beta$. Sum is taken in the current domain — naturally linear, fast, and low-glitch.

Applications

Problem. Receiver gets serial data with no clock. Local reference clock is at the right *frequency* but unknown *phase*.

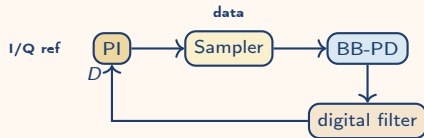
PI's job. Slide the sampling clock until it lands in the *centre of the data eye* — and keep it there as the link drifts.

$$\phi_{samp} = \frac{1}{2}(\phi_{rise} + \phi_{fall})$$



Why a PI?

Resolution of $360^\circ/2^N$ at multi-GHz rates with no PLL in the loop. A bang-bang phase detector dithers the PI code ± 1 LSB at lock — the entire CDR loop is digital after the sampler.



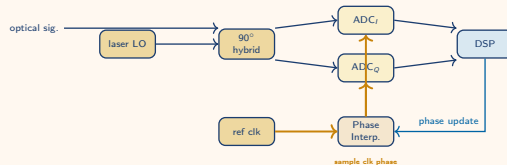
Setting. 100–800 Gb/s coherent links: optical front end delivers I/Q baseband into ADCs at 50 GS/s+.

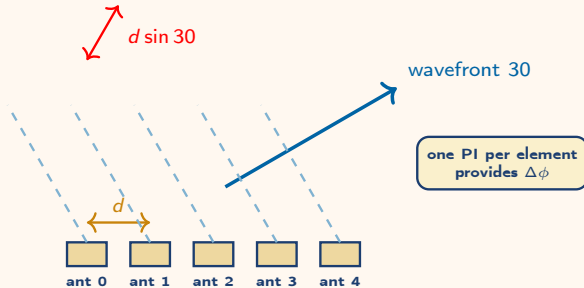
PI's role:

- sets the *ADC sample-clock phase* per lane
- tracks slow TX/RX symbol-clock drift, no analog retuning
- per-lane codes equalise inter-lane skew

Resolution matters

Clock jitter degrades SNR for 16/64-QAM. A 7-bit PI gives $\Delta\phi \approx 2.8^\circ \approx 80$ fs at 100 GHz.





$$\begin{aligned} \text{progressive delay } \Delta t &= (d \sin 30)/c \\ \Rightarrow \text{progressive phase } \Delta\phi &= \frac{2\pi d}{\lambda} \sin 30 \end{aligned}$$

Idea. Feed each antenna the *same signal* but with a progressive phase shift $\Delta\phi$ between neighbours. The array radiates a beam at angle

$$\sin \theta = \frac{\lambda}{2\pi d} \Delta\phi.$$

Change $\Delta\phi$ electronically $\rightarrow$ steer θ with no moving parts.

Why a PI per element?

- Continuous, monotonic phase
- Per-element skew calibration
- Same hardware across the band — only $\Delta\phi$ changes

Array equation (d spacing, λ wavelength). Differentiate to map small $\Delta\phi$ to small $\Delta\theta$:

$$\Delta\phi = \frac{2\pi d}{\lambda} \sin \theta \implies \boxed{\Delta\theta = \frac{\lambda}{2\pi d \cos \theta} \Delta\phi_{\min}}$$

For an N -bit PI ($\Delta\phi_{\min} = 2\pi/2^N$) at the typical half-wavelength spacing $d = \lambda/2$:
 $\Delta\theta = (1/\cos \theta) \cdot 2^{-(N-1)}$ rad.

N	$\Delta\phi_{\min}$	$\Delta\theta_{@0^\circ}$	$\Delta\theta_{@60^\circ}$
4	22.5°	7.2°	14.3°
6	5.6°	1.8°	3.6°
8	1.4°	0.45°	0.90°

Two asymmetries

$1/\cos \theta$ blows up near endfire — same LSB buys coarser steps off-broadside.
 And PI *nonlinearity* (DNL/INL) sets the worst-case step, not the average.

- A PI is a **weighted sum of two reference phases**; the weight ratio sets $\phi = \arctan(\beta/\alpha)$, the frequency is unchanged.
- Geometry is just the phasor tip sliding inside an I/Q quadrant; coarse = quadrant select, fine = N -bit weight DAC.
- Realised as two CML pairs sharing a load — current-domain summation, no analog multiplier.
- **SerDes / optical**: sets the data sampling clock phase in fully digital CDR loops at GHz+ rates.
- **Phased arrays**: sets per-element progressive phase; $\Delta\theta = \frac{\lambda}{2\pi d \cos\theta} \Delta\phi$ ties phase resolution to angular resolution.
- Bit-count of the PI is the lever shared by every application — more bits buy finer eye-centering, lower DSP jitter, finer beam steps.

Questions?